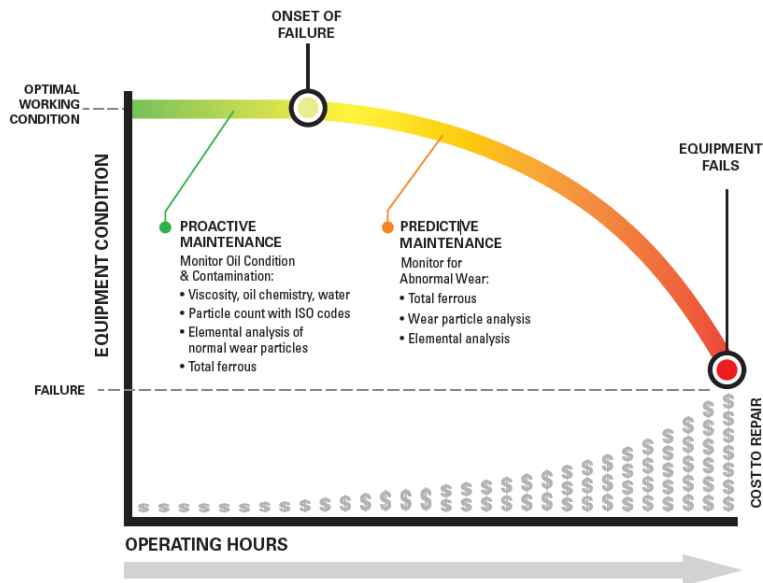


MiniLab Series for Industrial Plants

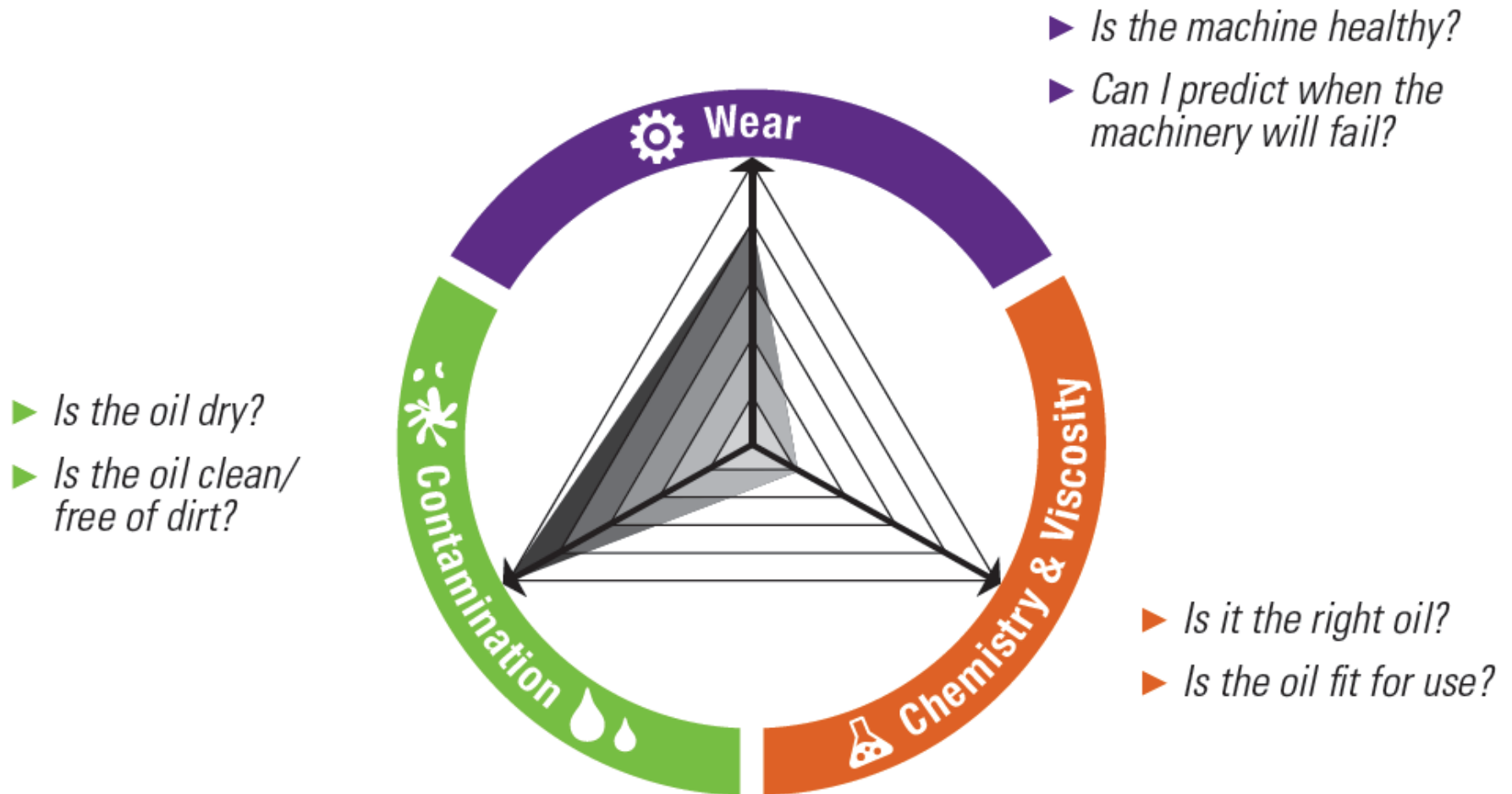


In service oil analysis for Machine Condition Monitoring



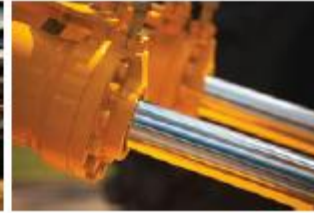
- Early identification and trending of machinery wear and failure modes allows maintenance before catastrophic failure occurs.
- Oil analysis complements vibration, thermography and other predictive technologies
- Increases confidence in decision to remove machinery from service.

Oil Analysis answers the critical questions ...



Trivector graph of machinery health

Typical limits for machinery types



MEASUREMENT	TURBINES	GEAR BOXES	HYDRAULICS	ENGINES
 Particle Count / ISO Code	< 18/14/12	< 19/16/13	< 15/13/11	
 Water	< 100 ppm	< 1,000 ppm	< 150 ppm	< 2,500 ppm
 Glycol				0.1 % max.
 Fuel dilution				5% max.
 Viscosity	+15%/-10 % of nominal ISO	+15%/-10 % of nominal ISO	+15%/-10 % of nominal ISO	+20 % to -10 % of nominal SAE
 TAN	Baseline + 0.1 mg KOH/g max.	Baseline + 1.0 mg KOH/g max.	Baseline + 1.0 mg KOH/g max.	
 TBN				20% of Baseline mg KOH/g min.
 Sodium				< 40 ppm
 Boron				< 20 ppm
 Aluminum, Chromium, Tin				< 15 ppm
 Iron, Lead				< 100 ppm
 Silicon				< 10 ppm
 Copper	< 30 ppm			< 40 ppm
 Zinc	< 2 ppm			

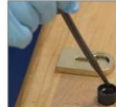
Key Analysis Parameters of the MiniLab 153

	PARAMETER	Elemental	Particle Count and Ferrous	Viscosity	Chemical
Contamination 	Particle count and ISO codes		✓		
	Non-metallic particle count, distribution and images		✓		
	Boron, Calcium, Sodium, Lithium, and Potassium	✓			
	Water				✓
Chemistry 	Viscosity			✓	
	Total Acid Number (TAN)				✓
	Oxidation				✓
	Copper, Chromium, Boron, Magnesium, Calcium, Barium, Zinc, Silicon, Sodium, Molybdenum, and Phosphorus	✓			
Wear 	Wear particle images, counts and distribution		✓		
	Total Ferrous content, ppm		✓		
	Ferrous particle count and size distribution		✓		
	Copper, Silver, Chromium, Titanium, Aluminum, Silicon, Magnesium, Nickel, Zinc, Iron, Manganese, Lead, Tin, Molybdenum, Cadmium, and Vanadium	✓			

4 simple tests to quantify machinery health

MiniLab 153

1



Elemental
Analysis



2



Particle
Count and
Ferrous



3



Viscosity

4

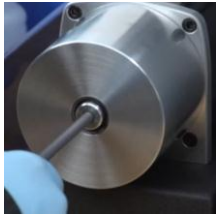


Chemical
Analysis

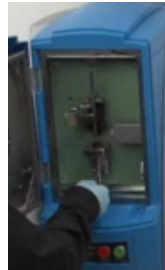
Four simple tests - Elemental Analysis

Test 1: Elemental analysis

- Sharpen electrode and install with disk
- Pipette oil sample into cup
- Run sample



Sharpen
electrode



Insert
electrode
and disk



Load sample into cup



Test

Four simple tests - Particle Count and Ferrous

Test 2: Particle Count and Wear

- Dilute sample if viscosity >320 cSt
- Shake 30 seconds by hand
- 30 seconds in ultrasound to remove bubbles
- 3 minutes test through LNF particle analyzer



Dilute
Sample



Shake
by hand



Ultrasound
(deaeration)



Test

Four simple tests - Viscosity & Chemistry



Test 3: Viscosity

- Pipette oil sample, drop 60 uL into viscometer with positive displacement pipette
- Not homogenized

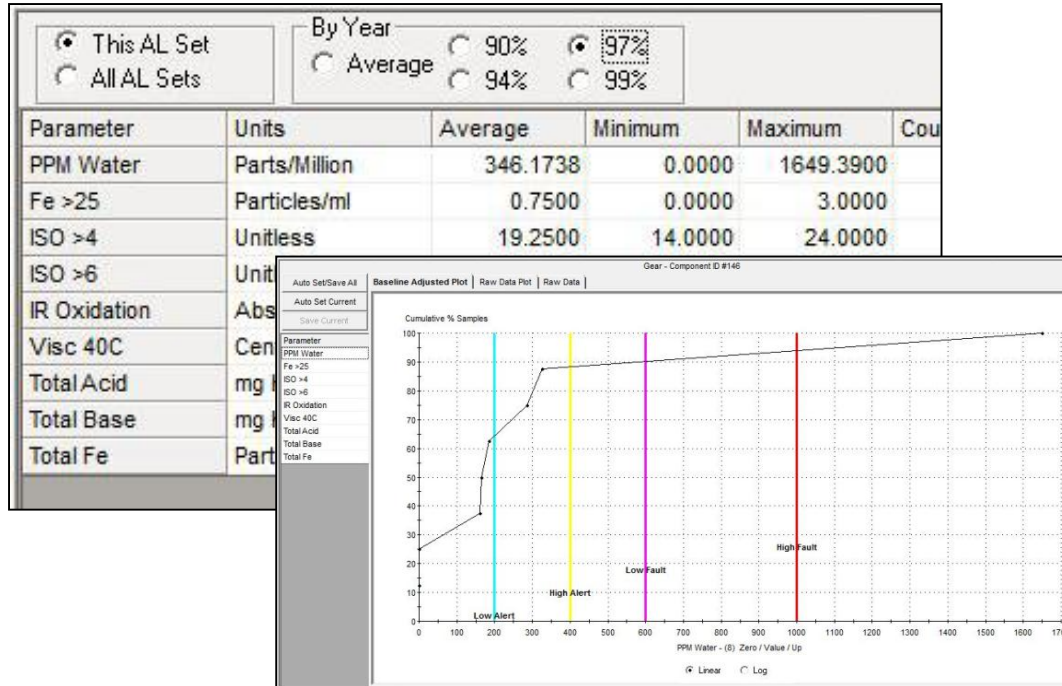


Test 4: Chemistry and Water

Homogenize sample, pipette 2 drops onto FluidScan

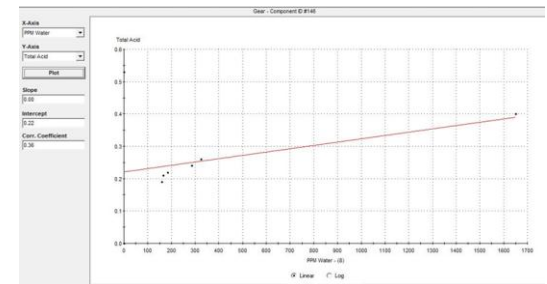
- Homogenization finely disperses free water so it can be accurately measured by the FluidScan

OilView Alarm analytics



- Review Alarm history by individual component or class of components
- Adjust initial alarm limits using the OilView cumulative probability distribution analysis capability

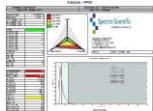
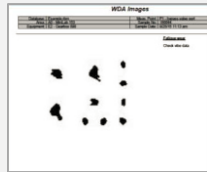
- Analyze correlations between Analysis Parameters for insight into root cause



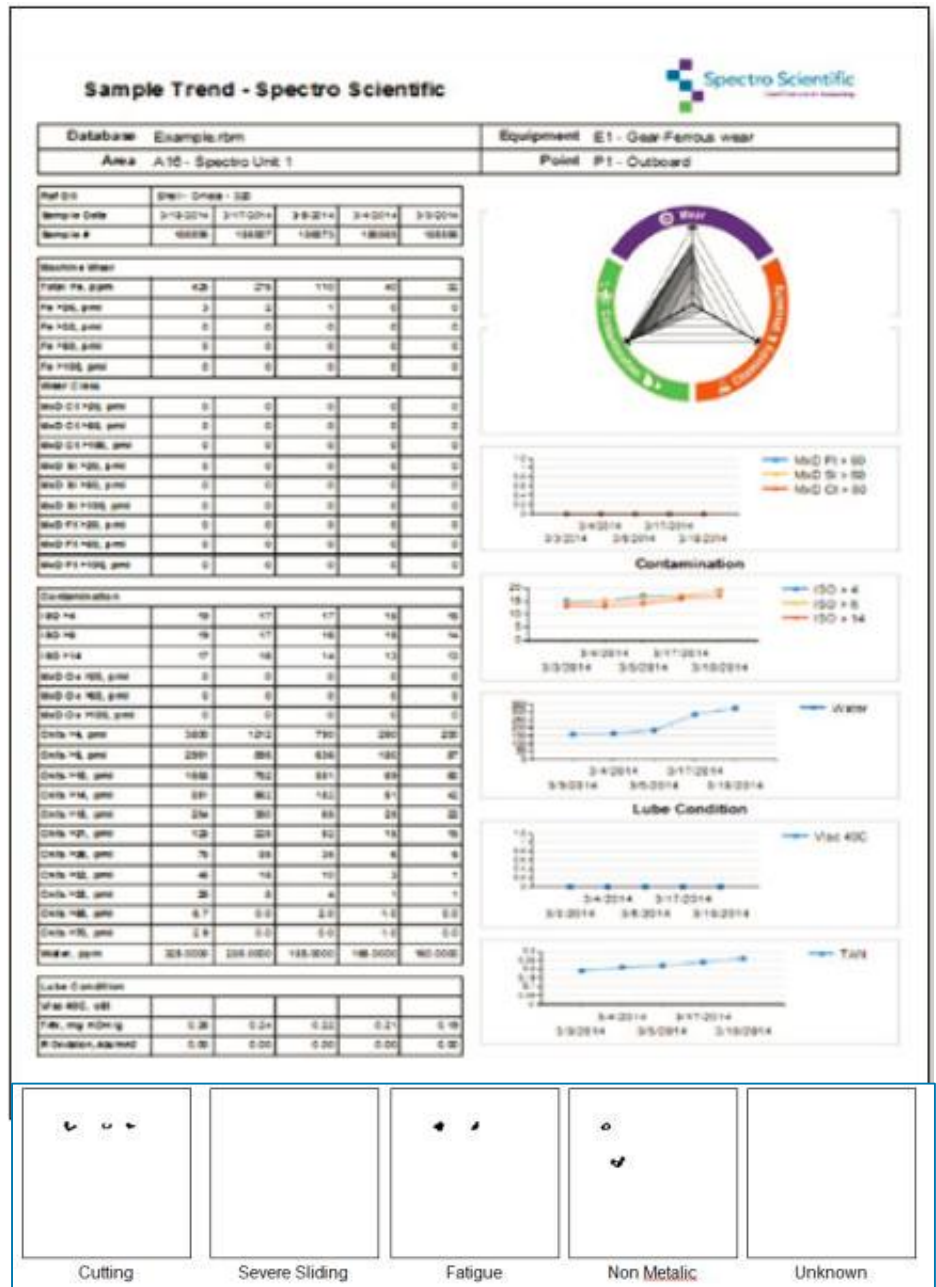
Trivector Reports

Trend Report with automatic import of Wear Image ‘thumbnails’

Optional WDA Module imports images from microscope camera or from LNF export file
-Export all images or filter by class



All traditional OilView reports still available



Performance verification with standard fluids

Total Ferrous and Ferrous Particle Count

LNF-545 Ferrous Validation Standard for Magnetometer

- Total Ferrous Content, nominal 100 ppm
- Large Ferrous particle count, nominal 42 um particles

Particle count

- LNF-509 calibration check fluid



Elemental

- CS-24 standard

Fluidscan IR check fluid

- FL311 FluidScan Check Fluid

Viscosity standards

- PV1025 Visc 30 standard
- PV1026 Visc120 standard

NIST traceable standards and an ASTM method for instrumentation

Minilab series configurations



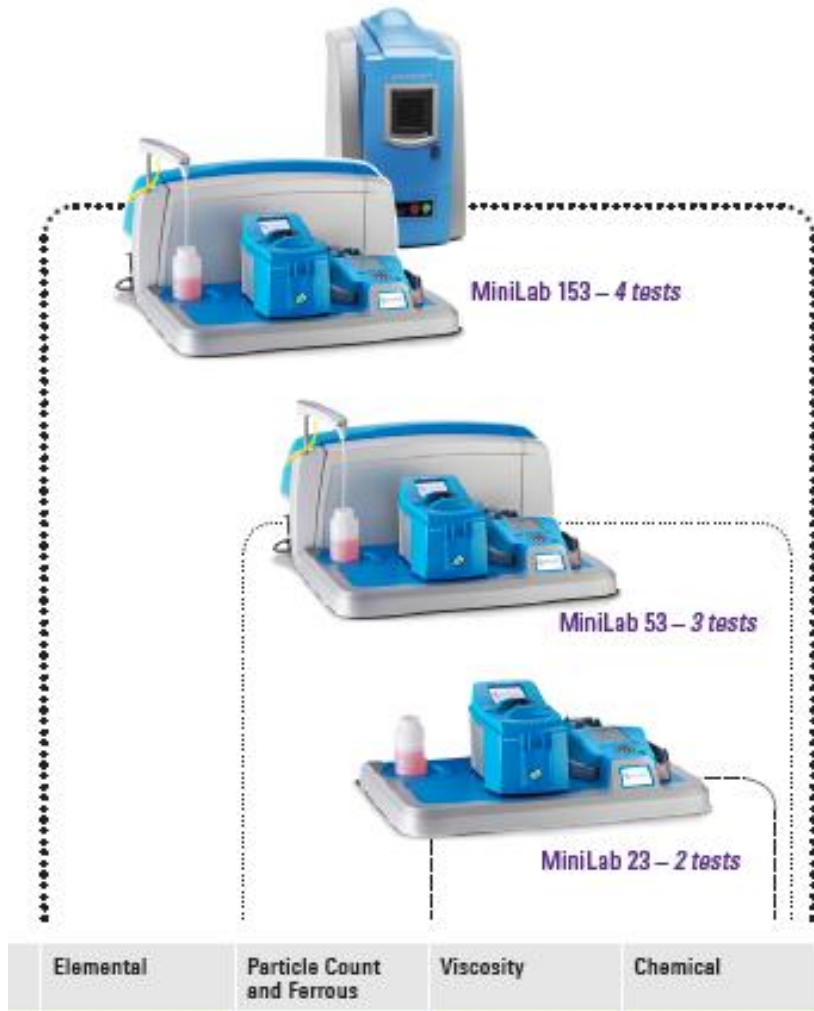
Minilab 153



Minilab 53

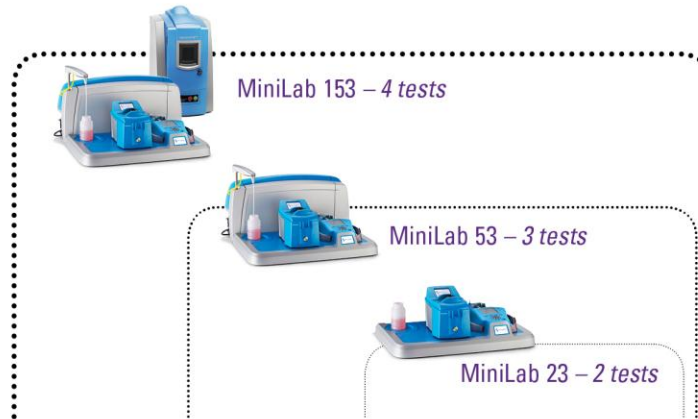


Minilab 23



Three models with increasing capability

MiniLab parameters



	PARAMETER	Elemental	Particle Count and Ferrous	Viscosity	Chemical
Contamination 	Particle count and ISO codes		✓		
	Non-metallic particle count, distribution and images		✓		
	Boron, Calcium, Sodium, Lithium, and Potassium	✓			
	Water				✓
Chemistry 	Viscosity			✓	
	Total Acid Number (TAN)				✓
	Oxidation				✓
	Copper, Chromium, Boron, Magnesium, Calcium, Barium, Zinc, Silicon, Sodium, Molybdenum, and Phosphorus	✓			
Wear 	Wear particle images, counts and distribution		✓		
	Total Ferrous content, ppm		✓		
	Ferrous particle count and size distribution		✓		
	Copper, Silver, Chromium, Titanium, Aluminum, Silicon, Magnesium, Nickel, Zinc, Iron, Manganese, Lead, Tin, Molybdenum, Cadmium, and Vanadium	✓			

A solution for every oil program with a clear upgrade path

Excellent repeatability of measurements for trending

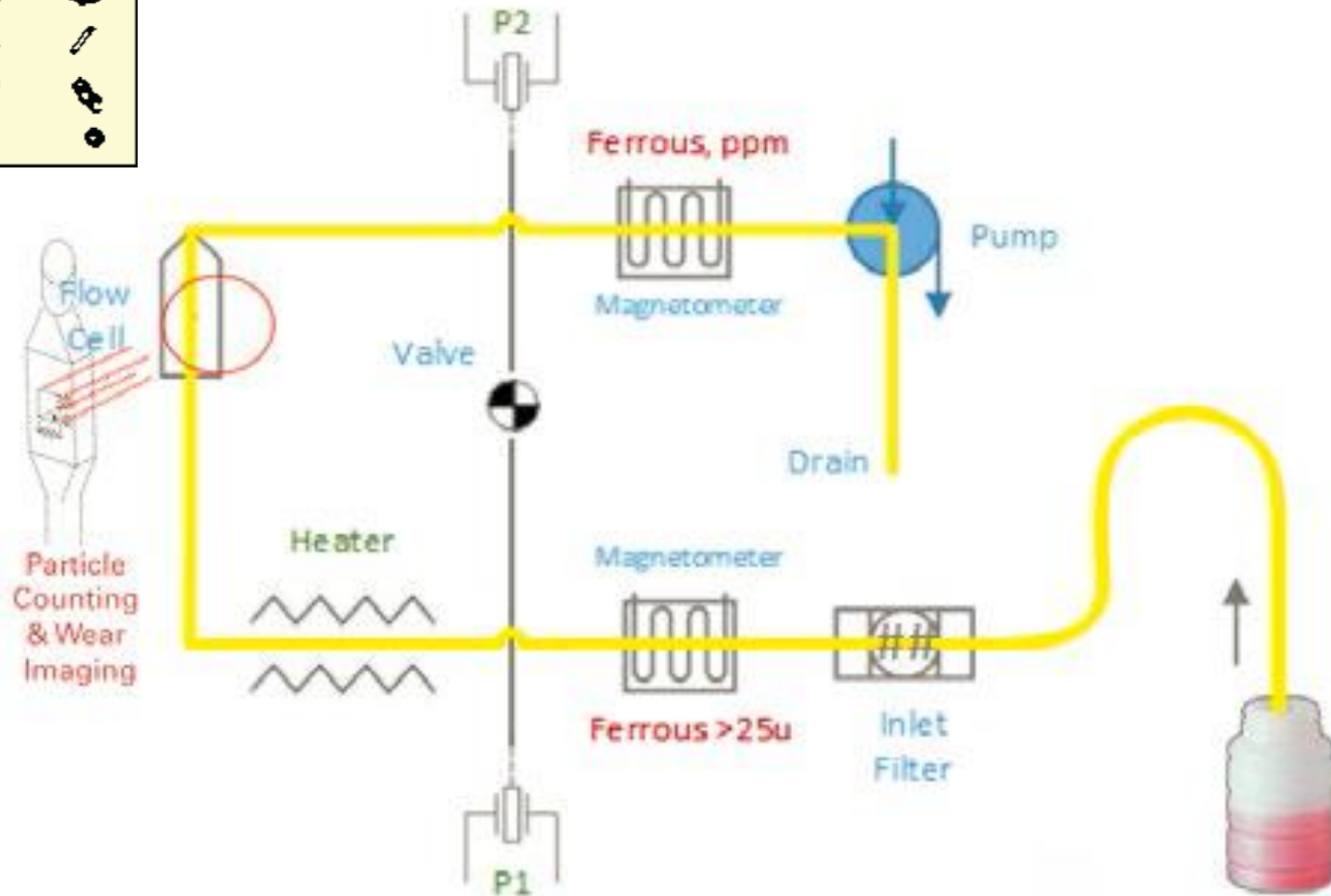
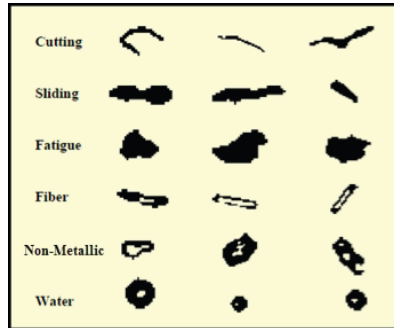
	Analytical Range	Repeatability
Particle Count	4 um-100um	± 6% RSD
Total Ferrous	10 - 2,000 ppm	± 5% RSD
Ferrous Particle Count	25-100 um	± 5% RSD
Viscosity (Low)	1-320 cSt	± 3% RSD
Viscosity (High)	320-700 cSt	± 5% RSD
TAN	0-6 mg KOH/g	± 3% RSD
Oxidation	0-3 abs/mm2	± 3% RSD
Dissolved water	200 ppm-saturation	± 6% RSD
Free water	0.1-6.5% (1,000-65,000 ppm)	± 25% RSD
Elemental	Varies with element	Varies with element

Lab report vs. OilView® TriVector™ report

	Typical lab report	Trivector report
Elemental analysis	Yes	
Particle count & ISO codes		
Viscosity,40C		
Water		
TAN/TBN		
Oxidation		
Total Ferrous	PQ/DR ferrography	Fe in ppm
Large Fe count & distribution	DR ferrography	Fe distribution >20um
Analytical ferrography	Optional	Automated classification
Wear images	X	Yes
Wear classification (cutting, sliding, fatigue)	X	Trendable data
Non-metallic count & distribution	X	Trendable data

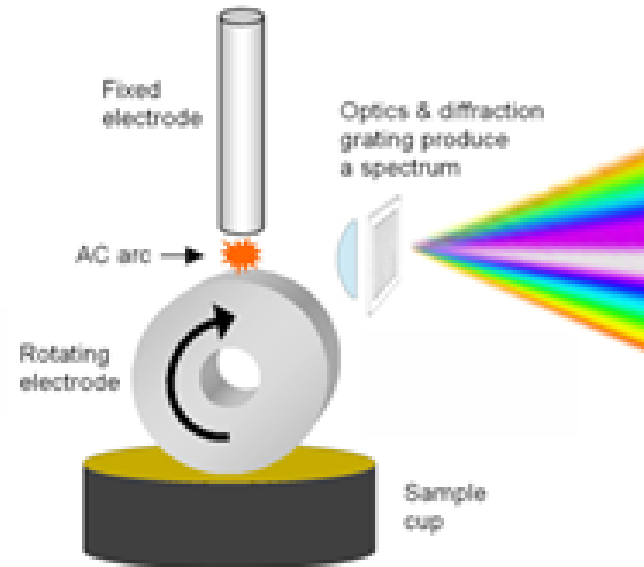
Simplified for on-site oil analysis - automated ferrography for the non-expert

How does LNF particle analyzer work

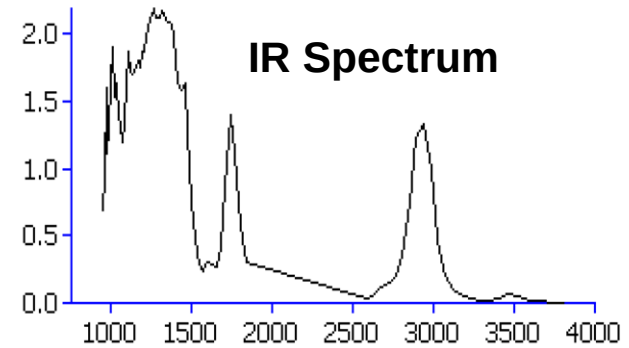
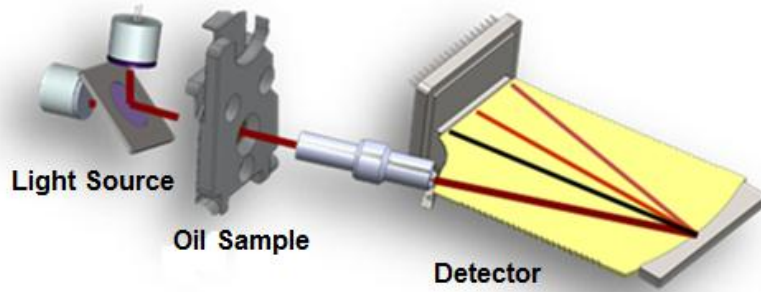


How does the RDE Elemental analyzer work

- Oil sample is loaded into cup. Rotating disk carries oil into AC arc
- Emission lines emitted representative of specific elements
- Spectrum captured with CCD detectors

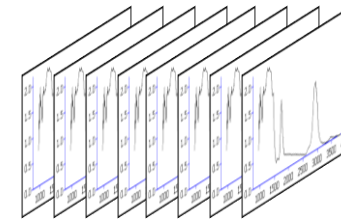


How does Fluidscan® work



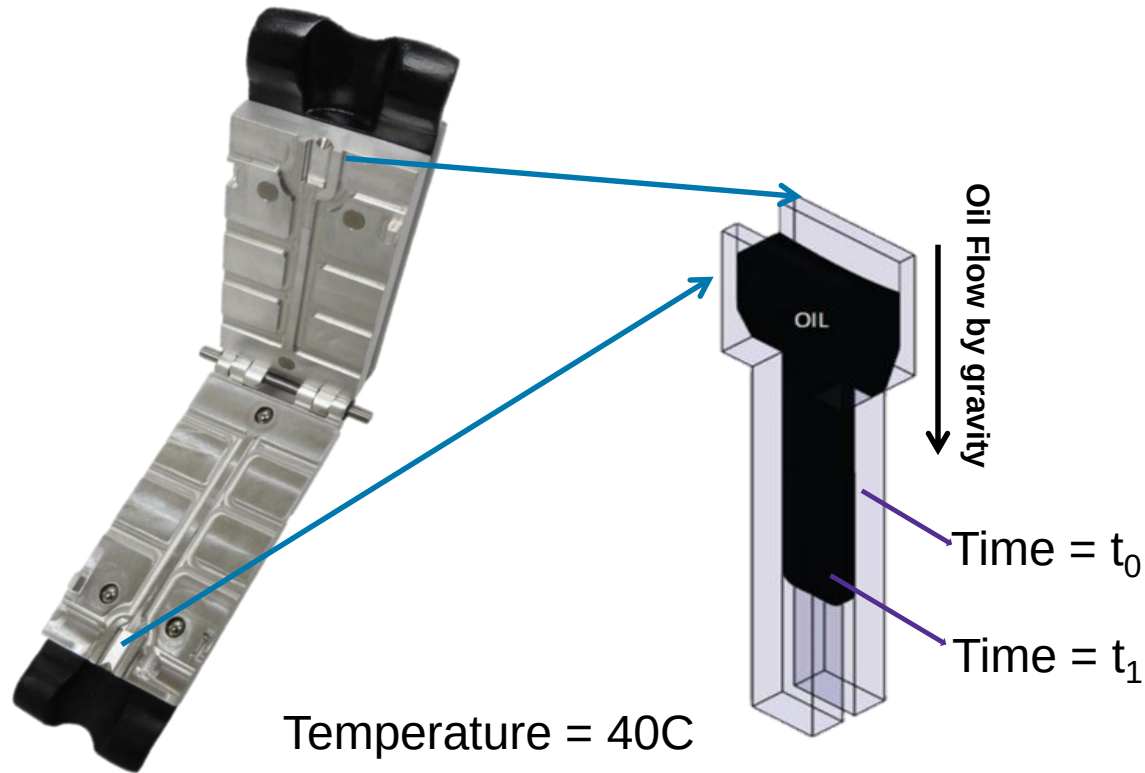
Measure Fluid » Results	
Sample ID: sample 1	
CAT NGEO EL250 SAE 40	
21 May 2014 01:31:15	
AW Additive	105 %
Glycol	0.5 %
Nitration	1.4 abs/0.1
Oxidation	12.4 abs/0.1
Soot	0.00 %wt
Sulfation	18.1 abs/0.1
TAN	0.73 mgKOH/L
TBN	5.0 mgKOH/L
Water	102 ppm

Measured fluid properties



Built-in Fluid Reference Library

How does the viscometer work



Elapsed time for sample to travel under gravity down 40C heated cell is measured and the Kinematic Viscosity calculated

$$\text{Viscosity} = A * (t_1 - t_0) + B$$

*A and B are calibration coefficients

MiniLab 153 analyses



CHEMISTRY

The Infrared Spectrometer measures TAN (Total Acid Number), Water and Oxidation for machinery oils and hydraulics.

For engine oil it measures oxidation, nitration, sulfation, total base number (TBN), soot and water



ELEMENTAL ANALYSIS

The Optical Emission Spectrometer measures 23 elements: wear metals from mechanical components, elements in oil additives and contamination.



VISCOSITY

The Viscometer measures 40C kinematic viscosity. It also calculates 100C viscosity using the Viscosity Index (VI).



PARTICLE COUNT

The Particle Counter, Ferrous Monitor and Wear Classifier provides:

- Particle Count & ISO Contamination Codes
- Total Ferrous content and Ferrous particle count
- Wear particle count and classification

The **MiniLab 153** delivers comprehensive on-site oil analysis with:

- 4 simple tests and results in < 20 minutes
- Lab-quality analysis results in your plant
- Simplified workflow for the non-expert user
- Trivector reports with immediate actionable results
- Full integration to AMS OilView software





Spectro Scientific

Confidence in knowing